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Routing and Scheduling in Tramp and Industrial Shipping

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Abstract In this chapter, we describe some important routing and scheduling problems in tramp and industrial shipping. We present both a vessel flow and a set partitioning model for what we denote as the basic version of the Tramp Ship Routing and Scheduling Problem (TSRSP) and we discuss how these models can be adapted to the industrial shipping case. Furthermore, we introduce and formulate several practical extensions to the TSRSP, including the integration of bunker planning (i.e., refueling of the vessels), as well as speed and sailing path optimization in the case of Emission Control Areas (ECAs). We hope this chapter will stimulate and help researchers and students interested in maritime routing problems.

Keywords Maritime transportation; Ship routing and scheduling; Tramp shipping; Industrial shipping; Speed optimization; Bunker optimization

1.1 Introduction

In 2021 around 11,000 million tons were transported at sea (UNCTAD, 2022), which illustrates the importance of maritime transportation for international trade. For ship operators, proper planning of the vessel fleet is crucial, as both the revenues and costs involved are normally very high. Despite this, the maritime sector has been slow to adopt automated planning approaches compared to many other industries (Ksciuk et al., 2023). Nonetheless, optimization-based decision support systems now play a major role also for several ship operators.

Following the categorization in previous survey papers on routing and scheduling in maritime transportation (Ronen, 1983, 1993; Christiansen et al., 2004, 2013), we distinguish between the modes *industrial*, *tramp* and *liner shipping*. In industrial shipping, the cargo owner (or shipper) controls the vessel fleet and aims at transporting its own cargoes at minimum costs. In tramp shipping, the vessels follow the available cargoes, where the cargoes can both be mandatory (e.g., originating from long-term contracts) and optional (from the spot market). A tramp ship operator tries to maximize the profit by selecting the most profitable spot cargoes while at the same time transporting all the contracted mandatory cargoes. In liner shipping, in which container shipping is the dominant segment, the vessels follow given routes according to a published schedule, usually on a weekly basis.

In this chapter, we intend to provide an overview of some important planning problems in maritime transportation. The main emphasis is on the modeling, although some solution methods are briefly discussed at a general level. We focus on routing problems in the first two modes of maritime transportation, i.e., industrial and tramp shipping. As such, this chapter can be considered as a follow-up on Christiansen and Fagerholt (2014). The industrial and tramp shipping modes include most of what is referred to as bulk shipping, i.e., transportation of wet bulk products (oil, chemicals and oil products) and dry bulk products (iron ore, grain, coal, bauxite/alumina, and phosphate). These shipping segments include more than 50% of the total seaborne volumes (UNCTAD, 2022). For the reader that is interested in liner shipping, we can refer to the surveys and studies by Meng et al. (2014), Brouer et al. (2014), Tran and Haasis (2015), Brouer et al. (2017), Christiansen et al. (2020) and Dulebenets et al. (2021).

This chapter is outlined as follows. We start in Section 1.2 by introducing what we refer to as the basic version of the Tramp Ship Routing and Scheduling Problem (TSRSP), for which we present both a vessel (or vehicle) flow and a set partitioning model. We briefly discuss some main solution methods for this problem and we also explain how these models can easily be adapted to the industrial shipping version. Next, in Section 1.3, we present some important practical extensions to the TSRSP (with mathematical models), including the integration of bunker and speed optimization, as well as routing and speed optimization with Emission Control Areas (ECAs). Finally, Section 1.4 draws some concluding remarks.

1.2 Tramp ship routing and scheduling: Basic version

In the basic version of the Tramp Ship Routing and Scheduling Problem (TSRSP), we consider a ship operator that controls a given heterogeneous fleet of vessels to be used for the following planning horizon (typically varying from a few weeks in short-sea shipping to a couple of months in deep-sea shipping applications). Each vessel in the fleet has its given characteristics, e.g., cargo capacity, fuel (bunker) consumption, and sailing speed. Since this planning problem in most cases is "open-ended" (i.e., the vessels are operating around the clock without having any depots), each vessel has a given origin, which can be both a port or a point at sea, and a time when the vessel becomes available for new transport tasks in the beginning of the planning horizon. We assume in the following that the vessels are empty when they become available at the beginning of the planning horizon.

We consider a given set of cargoes to be transported by the vessel fleet over the planning horizon. We distinguish between mandatory contract cargoes, for which the ship operator has already agreed to transport. Since a tramp ship operator also operates in a spot market, we assume there is also a set of optional spot cargoes, which can be transported if the ship operator

finds it beneficial (i.e., it improves the profit) and has sufficient fleet capacity to do so. In any case, each cargo consists of a given amount of product(s) to be picked up at a given loading port, transported, and discharged at a given delivery port. We assume that several cargoes may be transported simultaneously in a ship, but no allocation of the products into different compartments is considered. The product(s) transported is either assumed to be the same for different cargoes or the allocation can be solved by the people responsible for stowage, or as a separate planning problem. There is a time window in which the loading (pickup) of the cargo must start, and there may also be a discharging (delivery) time window. Furthermore, each cargo generates a given revenue when being transported.

The TSRSP consists of determining which spot cargoes to transport as well as the routes and schedules for the vessels in the fleet such that the profit (gross margin), i.e., the revenue of all transported cargoes minus the variable sailing costs (fuel, port, and canal costs), is maximized. The solution must ensure that all mandatory contract cargoes are serviced and that all time windows and vessel capacity constraints are respected. Furthermore, the TSRSP also may include the possibility for the ship operator to *relet* contract cargoes, which corresponds to chartering a vessel (at a cost) from the spot market to transport a cargo. In this case, the TSRSP also includes the decision about which cargoes to relet.

1.2.1 Vessel flow model

It should be noted that the notation used in this chapter differs slightly from what is used in several other chapters in this book. This is done mainly to be somewhat consistent with previous models in the literature on tramp ship routing and scheduling. From this follows for example that we mainly use capital letters for parameters and small letters for variables.

Let the fleet of vessels be represented by the set K , where each vessel is indexed by k . Assume there are n cargoes available in the following planning horizon, where each cargo is indexed by i . The sets of pickup (load) and delivery (discharge) nodes are given by $V^P = \{1, \dots, n\}$ and $V^D = \{n+1, \dots, 2n\}$, respectively. This allows each cargo i to be represented by a pickup node $i \in V^P$ and a corresponding delivery node $n+i \in V^D$. As such, the set of all cargo-related nodes is represented as $V = V^P \cup V^D$. The set of pickup nodes V^P is partitioned into $V^P = V^C \cup V^O$, where V^C and V^O are disjoint sets that denote the mandatory contracted and optional spot cargoes, respectively. Each vessel k has a node of origin $o(k)$, corresponding to the initial position of the vessel. For modeling purposes, each vessel is also assigned an artificial destination node $d(k)$, which physically corresponds to the vessel's last planned delivery node. Due to e.g., cargo capacity, cargo type compatibility, time window constraints, and port draft restrictions, some vessels in K and nodes in V may be incompatible. Thus, the set of nodes a vessel k may visit is represented as $V_k \subseteq V \cup \{o(k), d(k)\}$. (V_k, A_k) defines a graph for each vessel k , where the set of arcs $A_k \subseteq \{(i, j) | i, j \in V_k, i \neq j\}$ represents all arcs traversable by the given vessel. The pickup and delivery nodes compatible with vessel k are defined as $V_k^P = V^P \cap V_k$ and $V_k^D = V^D \cap V_k$, respectively.

Let T_{ijk}^S be the sailing time between nodes i and j for vessel k , while the time in port required to load or discharge the cargo at node i with vessel k is defined as T_{ik}^P . The variable voyage cost C_{ijk} accounts for the costs of visiting node i (port costs) and sailing between nodes i and j for vessel k (including any canal costs). The charter cost of servicing a contracted cargo i by a spot vessel (i.e., reletting the cargo) is denoted by the parameter C_i^S . Cargo $i \in V^P$ has a given quantity L_i to be transported which generates a given revenue R_i . For each node $i \in V_k$, there is

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a time window $[T_{ik}, \bar{T}_{ik}]$ for the pickup/delivery time, defined by the earliest time T_{ik} and the latest time $\bar{T}_{ik}$ for start of service. Note that these time windows are tightened by making them dependent of vessel k (e.g., by utilizing the position and time the vessel becomes available). The cargo carrying capacity of vessel k is given by Q_k .

The binary variable x_{ijk} is assigned the value 1 if vessel k traverses arc $(i, j) \in A_k$, and 0 otherwise. The variable t_{ik} denotes the time vessel k begins service at node i . The cargo load on board a vessel k when it leaves node i is represented by the variable l_{ik} . The binary variable z_i is assigned the value 1 if contracted cargo i is serviced by a spot vessel, and 0 otherwise. The binary variable y_i is assigned a value of 1 if optional spot cargo i is serviced, and 0 otherwise.

All the notation is summarized in Table 1.2.1.

Indices	
k	Vessel index
i, j	Port node index
Parameters	
T_{ijk}^S	Sailing time from node i directly to node j for vessel k
T_{ik}^P	Time required to load or discharge at node i with vessel k
C_{ijk}	Cost of servicing node i and sailing directly from node i to node j with vessel k
C_i^S	Cost of servicing cargo i with a spot vessel
R_i	Revenue generated from transporting cargo i
L_i	Quantity of cargo i
$\bar{T}_{ik}$	The latest time at which vessel k may begin its service at node i
T_{ik}	The earliest time at which vessel k may begin its service at node i
Q_k	Cargo carrying capacity of vessel k
n	Number of available cargoes
Sets	
K	Set of vessels in the fleet
V	Set of all cargo-related nodes
V^C	Set of pickup nodes for the mandatory contracted cargoes
V^O	Set of pickup nodes for the optional spot cargoes
V_k	Set of nodes that can be serviced by vessel k , including the origin node $o(k)$ and the artificial destination node $d(k)$
V_k^P	Set of pickup nodes that vessel k may service
V_k^D	Set of delivery nodes that vessel k may service
A_k	Set of all arcs traversable by vessel k
Decision variables	
x_{ijk}	1 if vessel k sails directly between nodes i and j , 0 otherwise
t_{ik}	Time vessel k begins service at node i
l_{ik}	Cargo load on board vessel k just after completing service at node i
y_i	1 if optional spot cargo i is transported, 0 otherwise
z_i	1 if cargo i is serviced by a vessel from the spot market, 0 otherwise

Table 1.2.1: Notation for the TSRSP

Objective Function

$$\max \sum_{i \in V^C} R_i + \sum_{i \in V^O} R_i y_i - \sum_{k \in K} \sum_{(i,j) \in A_k} C_{ijk} x_{ijk} - \sum_{i \in V^C} C_i^S z_i \quad (1.1)$$

Objective function (1.1) maximizes the profit (or gross margin to be more precise). The two first terms specify the revenue generated for the contracted and optional cargoes transported, respectively. The third term specifies the variable voyage costs for the vessel fleet, while the last term represents the charter costs associated with servicing any contracted cargoes with spot vessels.

Network Flow Constraints

$$\sum_{k \in K} \sum_{j \in V_k} x_{ijk} + z_i = 1 \quad i \in V^C \quad (1.2)$$

$$\sum_{k \in K} \sum_{j \in V_k} x_{ijk} - y_i = 0 \quad i \in V^O \quad (1.3)$$

$$\sum_{j \in V_k} x_{o(k)jk} = 1 \quad k \in K \quad (1.4)$$

$$\sum_{j \in V_k} x_{ijk} - \sum_{j \in V_k} x_{jik} = 0 \quad k \in K, i \in V_k \setminus \{o(k), d(k)\} \quad (1.5)$$

$$\sum_{i \in V_k} x_{id(k)k} = 1 \quad k \in K \quad (1.6)$$

$$\sum_{j \in V_k} x_{ijk} - \sum_{j \in V_k} x_{n+i,jk} = 0 \quad k \in K, i \in V_k^P \quad (1.7)$$

Constraints (1.2) ensure that all contract cargoes are transported by a vessel in the fleet or by a spot vessel. In contrast, constraints (1.3) state that all spot cargoes must be transported by a vessel in the predefined fleet or not at all. Constraints (1.4) ensure that each vessel begins a voyage from its origin node and sails either first to a pickup node or directly to its destination node (meaning that the vessel is not used within the planning horizon). Constraints (1.5) denote the flow conservation, while constraints (1.6) ensure that all vessels end their routes in their destination nodes. Constraints (1.7) ensure that the same vessel services both the pickup and the cargo's corresponding delivery node.

Temporal Constraints

$$x_{ijk} (t_{ik} + T_{ik}^P + T_{ijk}^S - t_{jk}) \leq 0 \quad k \in K, (i, j) \in A_k \quad (1.8)$$

$$t_{ik} + T_{ik}^P + T_{i,n+i,k}^S - t_{n+i,k} \leq 0 \quad k \in K, i \in V_k^P \quad (1.9)$$

$$\underline{T}_{ik} \leq t_{ik} \leq \bar{T}_{ik} \quad k \in K, i \in V_k \quad (1.10)$$

The above constraints handle the temporal aspects of the problem. Constraints (1.8) state that if vessel k sails directly between nodes i and j , the time at which a vessel k begins service in node j must be greater than or equal to the time at which the vessel began servicing node i plus the sum of time spent in port and sailing at sea. These constraints are non-linear but can easily be linearized as follows: $t_{ik} + T_{ik}^P + T_{ijk}^S - t_{jk} - M_{ijk}(1 - x_{ijk}) \leq 0$, with an appropriately

chosen big-M value. The inequality signs of constraints (1.8) give vessels the option to wait before beginning their service at node j . Constraints (1.9) are precedence constraints forcing the pickup node to be serviced before its corresponding delivery node. Finally, constraints (1.10) ensure that the services at the nodes are done within the specified time windows.

Cargo Constraints

$$x_{ijk} (l_{ik} + L_j - l_{jk}) = 0 \quad k \in K, (i, j) \in A_k \quad (1.11)$$

$$|j \in V_k^P$$

$$x_{i,n+j,k} (l_{ik} - L_j - l_{n+j,k}) = 0 \quad k \in K, (i, n+j) \in A_k \quad (1.12)$$

$$|j \in V_k^P$$

$$l_{o(k),k} = 0 \quad k \in K \quad (1.13)$$

$$\sum_{j \in V_k} L_i x_{ijk} \leq l_{ik} \leq \sum_{j \in V_k} Q_k x_{ijk} \quad k \in K, i \in V_k^P \quad (1.14)$$

$$0 \leq l_{n+i,k} \leq \sum_{j \in V_k} (Q_k - L_i) x_{n+i,jk} \quad k \in K, i \in V_k^P \quad (1.15)$$

Constraints (1.11) state that if a vessel k sails from node i to a pickup node j , the cargo load on board the vessel just after completing service at node j must be equal to the cargo level on board the vessel when leaving node i adjusted for the quantity loaded at node j . Constraints (1.12) specify a similar cargo flow if vessel k sails directly from node i to delivery node $n+j$. Both constraints (1.11) and (1.12) are non-linear but can easily be linearized in a similar way as for constraints (1.8). Constraints (1.13) set the initial load on board each vessel to zero. Constraints (1.14) and (1.15) are the (strengthened) capacity constraints for pickup and delivery nodes, respectively.

Binary and Non-Negativity Constraints

$$x_{ijk} \in \{0, 1\} \quad k \in K, (i, j) \in A_k \quad (1.16)$$

$$y_i \in \{0, 1\} \quad i \in V^O \quad (1.17)$$

$$z_i \in \{0, 1\} \quad i \in V^C \quad (1.18)$$

$$t_{ik} \in \mathbb{R}^+ \quad k \in K, i \in V_k \quad (1.19)$$

$$l_{ik} \in \mathbb{R}^+ \quad k \in K, i \in V_k \quad (1.20)$$

Constraints (1.16) – (1.20) define the domains of all the variables.

Note that the basic version of the TSRSP modeled above is a version of the well-known Pickup and Delivery Problem with Time Windows (PDPTW), e.g., Dumas et al. (1991). However, there are also some notable differences. The PDPTW has a given depot where all vehicles start and end their routes, while in the TSRSP the vessels operate 24/7 with no depots and therefore have different initial positions and times they become available. Furthermore, in the TSRSP, there is a mix of mandatory and optional cargoes, while the PDPTW usually only has mandatory ones.

It should also be emphasized that the above model is for the *parcel cargo version* of the TSRSP, i.e., where a vessel can carry more than one cargo at the same time as long as the cargoes' total load is within the vessel's capacity. However, there are many shipping segments where the size of the cargo is adapted to the typical size of vessels, meaning that one usually has at most one cargo on board the vessel at the same time. This is often referred to as the *full shipload version* of the TSRSP. In this simpler case, the above model can be simplified by letting each cargo be represented by only one node, and the model then becomes a version of the Vehicle Routing Problem with Time Windows (VRPTW).

1.2.2 Set partitioning model

The basic TSRSP can also be formulated as a set partitioning model following a Dantzig-Wolfe decomposition, where each path corresponds to a feasible vessel route. Let R_v denote the set of feasible routes for vessel v . In this setting, a feasible route adheres to constraints (1.4) - (1.15). Furthermore, let the sets K , V , V_k , V^C , and V^O be defined as in the vessel flow model, denoting the set of vessels, cargo-related nodes, set of nodes vessel v may visit, contracted cargoes, optional spot cargoes, respectively. For each route $r \in R_k$ for vessel k there is a vessel-specific profit denoted P_{rk} . As in the vessel flow model, C_i^S denotes the cost of servicing the cargo at pickup node i with a spot vessel. A new binary parameter A_{irk} is introduced and set equal to 1 if vessel k services cargo i in route r , and 0 otherwise. The variable x_{rk} defines a binary variable equal to 1 if vessel k sails its route r , and 0 otherwise. Variables $y_i \in V^O$ and $z_i \in V^C$ are defined as before, denoting whether or not a spot cargo is serviced and whether or not a spot vessel is used to service an optional cargo, respectively.

The additional notation for the set partitioning model is summarized in Table 1.2.2.

Indices	
r	Route index
Parameters	
P_{rk}	Vessel specific profit generated by performing route r for vessel k
A_{irk}	1 if vessel k carries cargo i in route r , 0 otherwise
Sets	
R_k	Set of all routes for vessel k
Decision variables	
x_{rk}	1 if vessel k is chosen to sail route r , 0 otherwise

Table 1.2.2: Additional notation for the set partitioning model for the TSRSP

With the additional notation defined above, the basic TSRSP can be formulated with the following set partitioning model.

$$\max \sum_{k \in K} \sum_{r \in R_k} P_{rk} x_{rk} - \sum_{i \in V^C} C_i^S z_i \quad (1.21)$$

subject to

$$\sum_{k \in K} \sum_{r \in R_k} A_{irk} x_{rk} + z_i = 1 \quad i \in V^C \quad (1.22)$$

$$\sum_{k \in K} \sum_{r \in R_k} A_{irk} x_{rk} - y_i = 0 \quad i \in V^O \quad (1.23)$$

$$\sum_{r \in R_k} x_{rk} = 1 \quad k \in K \quad (1.24)$$

$$x_{rk} \in \{0, 1\} \quad k \in K, r \in R_k \quad (1.25)$$

$$y_i \in \{0, 1\} \quad i \in V^O \quad (1.26)$$

$$z_i \in \{0, 1\} \quad i \in V^C \quad (1.27)$$

The first term in the objective function (1.21) sums up the vessel-specific profit generated by each of the vessels in the fleet, while the second term denotes the cost of hiring spot vessels to service any contract cargoes. Constraints (1.22) state that each contracted cargo is either transported by a vessel in the fleet or by a spot vessel. Constraints (1.23) specify that optional cargoes may be serviced by a vessel in the fleet, while constraints (1.24) ensure that each vessel is assigned exactly one route. Constraints (1.25) – (1.27) confine the binary variables to their respective domains.

The above set partitioning model can easily be adapted to the case of industrial shipping, i.e., when there are only mandatory contracted cargoes. In that case, there are no spot cargoes so we no longer need constraints (1.23) and the y_i variables. For the same reason, the total revenues can be considered fixed, and the objective can instead be to minimize the variable sailing costs.

1.2.3 Solution methods

Many of the solution methods proposed for the TSRSP are based on solving a set partitioning model, as presented in Section 1.2.2. Many of these studies generate the set of feasible candidate routes a priori as input to the set partitioning model (e.g., Brown et al. (1987); Bausch et al. (1998); Fagerholt (2001)), while others generate the routes dynamically within a column generation and/or branch-and-price framework (e.g., Stålhané et al. (2012); Brønmo et al. (2010); Meng et al. (2015); Homsí et al. (2020)).

Different variants of the TSRSP have also been solved with a variety of heuristic solution methods. Examples of this include multi-start local search heuristics (Brønmo et al., 2007; Yamashita et al., 2019), (Adaptive) Large Neighborhood Search (Korsvik et al., 2011; Lee and Kim, 2015), and hybrid genetic search (Homsí et al., 2020)).

1.3 Extensions to the Tramp Ship Routing and Scheduling Problem

In many practical cases, the basic version of the TSRSP (as well as for the industrial shipping version) presented in Section 1.2 can be too simplified, as real problems often include both additional planning restrictions as well as planning flexibility. In the following, we present some of these important practical extensions.

1.3.1 Bunker optimization

In the above models for the TSRSP, it is assumed that bunkering (or refueling) of the vessels can be done in any port that is visited so it does not need to be explicitly planned for. However, this does not always have to be the case, and detours to bunkering ports might be needed. Furthermore, the bunker (fuel) price usually varies greatly among ports. It can therefore be important to jointly plan the routing of the vessels with the bunkering decisions (i.e., how much to bunker where). With the possible future introduction of alternative low- or zero-emission fuels, such as hydrogen, methanol, or ammonia, this might even become more important as there will most likely be fewer options of where to refuel due to limitations in infrastructure for these new fuel types.

The additional notation presented in the following mainly follows the bunker management extension devised by Vilhelmsen et al. (2014). Let B be the set of nodes (ports) offering the purchase of bunker. The first step is extending the base network (V_k, A_k) to allow vessel visits at bunker option nodes. Similarly, as for the cargo-related nodes, vessels and bunker option nodes may be incompatible. Thus, set $B_k \subseteq B$ denotes the bunker option nodes that vessel k may visit. The set of nodes V_k for vessel k , as defined in Section 1.2, is extended to include the nodes in B_k such that $\hat{V}_k = V_k \cup B_k$. Furthermore, the set of arcs A_k is extended by adding all arcs connecting nodes in V_k with nodes in B_k that are traversable by vessel k with respect to time and bunker. As such, the extended set of arcs is defined as $\hat{A}_k = A_k \cup A_k^B$, where A_k^B is the set of arcs for vessel k connecting bunker option nodes B_k to the nodes in $V_k \setminus \{d(k)\}$. Hence, the extended cargo-bunker network is thereby defined by $(\hat{V}_k, \hat{A}_k)$.

Let P_i^B denote the cost of purchasing one unit of bunker in bunker option node $i \in B$. To avoid negative end-of-horizon effects, we add a reward $\tilde{P}^B$ per unit for having bunker on board the vessels at the end of the planning horizon, which is calculated as the mean of all available bunker node prices. Furthermore, the bunker consumption at sea is denoted B_{ijk}^S for a vessel k traversing the arc $(i, j) \in \hat{A}_k$. The bunker consumed by vessel k in node i per unit of time is represented by B_{ik}^P . At the beginning of the planning horizon, each vessel has an initial amount of bunker stored, denoted by B_k^0 . Additionally, the bunker level on board each vessel k must be within a lower limit $\underline{B}_k$ and an upper limit $\bar{B}_k$ to ensure realistic and safe operation.

It should be noted that we now have to change the definition of the sailing cost parameter C_{ijk} . Now it should not include the fuel costs, as the cost of the bunker purchases is modeled separately. Hence, C_{ijk} now includes only the port costs in node i plus any canal costs between nodes i and j with vessel k .

The quantity of bunker purchased by vessel k at bunker option node i is given by the variable b_{ik} . The bunker load on board a vessel k when it leaves node i is represented by the variable l_{ik}^B .

The additional notation for the bunker optimization extension is summarized in Table 1.3.1.

The objective function (1.28) extends the objective function (1.1) by including terms relevant to the bunker extension. The first line is defined as before, subtracting variable sailing costs and spot vessel costs from the generated revenue. The first term on the second line describes the cost associated with purchasing the bunker, while the final term models the value of the bonus bunker remaining at the destination node. Note that if the bunker level at the destination is higher than at the origin, this term will positively contribute to the objective. Conversely, a lower bunker level at the destination than at the origin yields a negative contribution.

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Parameters	
P_i^B	Price of purchasing one unit of bunker at the bunker node i
$\tilde{P}^B$	Unit price of bonus bunker at the end of the planning horizon
B_{ijk}^S	Total bunker consumption for vessel k while sailing directly from node i to node j
B_{ik}^P	Port bunker consumption for vessel k while in node i
B_k^0	Initial bunker level on board vessel k
$\bar{B}_k$	Maximum bunker level for vessel k
$\underline{B}_k$	Minimum bunker level for vessel k
Sets	
B_k	Set of bunker nodes that vessel k may visit
$\hat{V}_k$	Extended set of nodes that can be visited by vessel k
$\hat{A}_k$	Extended set of arcs traversable by vessel k
Decision variables	
b_{ik}	Bunker quantity purchased by vessel k at bunker node i
l_{ik}^B	Bunker load on board vessel k just after completing service at node i

Table 1.3.1: Additional notation for the bunker optimization extension

Now, the model in Section 1.2 can be extended as follows to handle the bunker optimization extension.

$$\begin{aligned}
\max \quad & \sum_{i \in V^C} R_i + \sum_{i \in V^O} R_i y_i - \sum_{k \in K} \sum_{(i,j) \in \hat{A}_k} C_{ijk} x_{ijk} - \sum_{i \in V^C} C_i^S z_i \\
& - \sum_{k \in K} \sum_{i \in B_k} P_i^B b_{ik} + \sum_{k \in K} \tilde{P}^B \cdot (l_{d(k)k}^B - B_k^0)
\end{aligned} \tag{1.28}$$

We also get the following additional bunker constraints (1.29) – (1.34).

$$\begin{aligned}
x_{ijk} \left(l_{ik}^B - B_{ijk}^S - B_{jk}^P - l_{jk}^B \right) &= 0 & k \in K, (i,j) \in \hat{A}_k \\
& | j \in V_k \setminus \{d(k)\}
\end{aligned} \tag{1.29}$$

$$\begin{aligned}
x_{ijk} \left(l_{ik}^B - B_{ijk}^S + b_{jk} - l_{jk}^B \right) &= 0 & k \in K, (i,j) \in \hat{A}_k \\
& | j \in B_k
\end{aligned} \tag{1.30}$$

$$\begin{aligned}
x_{ijk} \left(l_{ik}^B - B_{ijk}^S - l_{jk}^B \right) &= 0 & k \in K, (i,j) \in \hat{A}_k \\
& | j \in \{d(k)\}
\end{aligned} \tag{1.31}$$

$$l_{o(k)k}^B = B_k^0 \quad k \in K \tag{1.32}$$

$$\sum_{j \in \hat{V}_k} \left(\underline{B}_k + B_{ijk}^S + B_{jk}^P \right) x_{ijk} \leq l_{ik}^B \leq \bar{B}_k \sum_{j \in \hat{V}_k} x_{ijk} \quad k \in K, i \in \hat{V}_k \tag{1.33}$$

$$b_{ik} \leq (\bar{B}_k - \underline{B}_k) \sum_{j \in \hat{V}_k} x_{ijk} \quad k \in K, i \in B_k \quad (1.34)$$

$$b_{ik} \in \mathbb{R}^+ \quad k \in K, i \in V_k \quad (1.35)$$

$$l_{ik}^b \in \mathbb{R}^+ \quad k \in K, i \in V_k \quad (1.36)$$

Constraints (1.29) state that if vessel k sails directly from node i to cargo node j , the bunker load on board just after completing service at node i minus the bunker consumption of sailing from node i to node j and consumed while servicing at node j must be equal to the bunker load on board just after completing service at node j . Constraints (1.30) do the same for bunker nodes at which there is no cargo handling. There is, however, bunker loading at node j which is added to the bunker load balance. Constraints (1.31) describe the bunker load balance during sailing to the destination node as there is no bunkering nor bunker consumption here. Constraints (1.29) – (1.31) are non-linear but can be linearized in a similar way as the cargo constraints (1.11) and (1.12). Constraints (1.32) specify the bunker level on board the vessel at the beginning of the planning period. Constraints (1.33) state that the bunker load on board each vessel just after completing its service at node i must be greater than or equal to the minimum bunker level of the vessel plus the bunker consumed during sailing and cargo handling in node j , and less than or equal to the total bunker capacity of the vessel. Constraints (1.34) ensure that the bunker purchased by a vessel at any bunker node is less than or equal to the difference between the vessel's minimum and maximum bunker load capacity. Constraints (1.35) – (1.36) specify the domains of the new bunker variables.

Vilhelmsen et al. (2014) address a joint (full shipload) tramp ship routing and bunker optimization problem which they solve with a column generation algorithm where the amounts of bunker to purchase are discretized. Furthermore, the column generation scheme is not embedded within a branch-and-bound search, which may lead to further non-optimality of the generated solutions. Meng et al. (2015) build on the work of Vilhelmsen et al. (2014) and propose a branch-and-price framework. Rather than discretizing the bunker amounts, they provide an algorithm that can obtain optimal bunkering decisions. However, Meng et al. (2015) restrict bunkering to take place only in cargo ports that are otherwise visited, i.e., no detours for bunkering are allowed, which significantly reduces the size of the underlying network. A recent contribution is by Omholt-Jensen et al. (2023), who solve the problem with a set partitioning model where feasible routes are generated a priori, and each of these is optimized with respect to the bunkering decisions. It can also be noted that the (tramp) ship routing and scheduling problem with bunker optimization has similarities with the Electric VRPTW, in which vehicle charging must be considered, e.g., (Schneider et al., 2014; Desaulniers et al., 2016).

1.3.2 Speed optimization

In the vessel flow model for the TSRSP introduced in Section 1.2, it is assumed that the vessels have a given sailing speed and fuel consumption. However, in reality, the fuel (energy) consumption, and hence the fuel cost, heavily depends on the sailing speed within the speed interval defined by a vessel k 's minimum speed $\underline{S}_k$ and maximum speed $\bar{S}_k$, e.g., (Psaraftis and Kontovas, 2014; Ronen, 1982). Hence, it can be beneficial to jointly determine the routes and the sailing speeds of the vessels in the TSRSP.

To model this extension, we define a new decision variable s_{ijk} to denote the speed for vessel k between nodes i and j . The cost for vessel k to sail the given distance D_{ij} between nodes i and

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j now becomes a (non-linear) function of this sailing speed, i.e., the parameter C_{ijk} becomes a function $C(s_{ijk})$. This function typically has the following shape: $C_{ijk} = g(p + s_{ijk}^q)$, where g , p and q are vessel-specific constants. In reality, the fuel consumption also depends on the load onboard the vessel (Psaraftis and Kontovas (2014)). However, for simplicity we assume here that this is not the case. The above additional notation is summarized in Table 1.3.2.

Parameters	
D_{ij}	Distance between nodes i and j
$\underline{S}_k$	Minimum sailing speed of vessel k
$\bar{S}_k$	Maximum sailing speed of vessel k
Decision variables	
s_{ijk}	Sailing speed for vessel k between nodes i and j
Functions	
$C(s_{ijk})$	Sailing cost for vessel k when using speed s_{ijk} between nodes i and j

Table 1.3.2: Additional notation for the speed optimization extension

When integrating the speed optimization into the TSRSP, the third term of the objective function (1.1) introduced in Section 1.2 now becomes as follows.

$$- \sum_{k \in K} \sum_{(i,j) \in A_k} C(s_{ijk}) x_{ijk} \quad (1.37)$$

Furthermore, introducing sailing speeds as decision variables affects the temporal constraints. Hence, constraints (1.8) and (1.9) can now be replaced by constraints (1.38) and (1.39), in addition to the new constraints (1.40) limiting the sailing speeds to be within the vessels' speed intervals.

$$x_{ijk} (t_{ik} + T_{ik}^P + D_{ij}/s_{ijk} - t_{jk}) \leq 0 \quad k \in K, (i, j) \in A_k \quad (1.38)$$

$$t_{ik} + T_{ik}^P + D_{i,n+i}/s_{i,n+i,k} - t_{n+i,k} \leq 0 \quad k \in K, i \in V_k^P \quad (1.39)$$

$$\underline{S}_k \leq s_{ijk} \leq \bar{S}_k \quad k \in K, (i, j) \in A_k \quad (1.40)$$

Note that the updated objective function becomes non-linear due to the non-linear fuel consumption function $C(s_{ijk})$. Furthermore, constraints (1.38) and (1.39) also include new non-linear terms since the sailing time now depends on the sailing speed.

The integration of speed optimization within ship routing and scheduling has been handled in many different ways. When solving the problem with a set partitioning model (as shown in Section 1.2.2) where the candidate routes are generated a priori, the sailing speeds can be optimized locally within each route (Brown et al., 1987; Fagerholt, 2001). Norstad et al. (2011) propose a polynomial algorithm for optimizing the sailing speeds along given routes and integrate this within a local search-based heuristic. Wen et al. (2016) also take into account that the fuel consumption is higher when the vessel is fully loaded than when sailing in ballast. They solve the problem with a branch-and-price algorithm with heuristic column generation.

1.3.3 Routing with emission control areas

Stricter limits on the maximum sulphur content in the fuel used by vessels are now imposed in Emission Control Areas (ECAs) along several coastlines. To comply with these regulations ship operators may have to switch to more expensive low-sulphur fuel when sailing within these ECAs. Since they are concerned about minimizing costs, optimal speed and routing decisions likely change because of this. Fagerholt et al. (2015) showed that in order to minimize sailing costs, ship operators could choose to use lower sailing speeds within the ECAs (due to the more expensive fuel and the non-linear shape of the fuel consumption function) and compensate for this by increasing the speeds outside the ECAs. Furthermore, it was shown that ship operators might in some cases even choose to sail longer distances to reduce the sailing within ECAs. As an example, consider the case in Figure 1, where a vessel is to sail between ports A and B, where both ports are located along a coastline within an ECA (as is the case e.g., along the coasts of North America). Then, it might be optimal not to choose the shortest sailing path along the coast, which is entirely within the ECA, but instead, choose an alternative longer path where most of the sailing distance is outside the ECA. Hence, an unintended consequence of the ECAs as a policy to reduce sulphur emissions along coastlines might be that vessels make detours so that other emissions (e.g., CO₂) actually increase, as shown by Fagerholt et al. (2015).

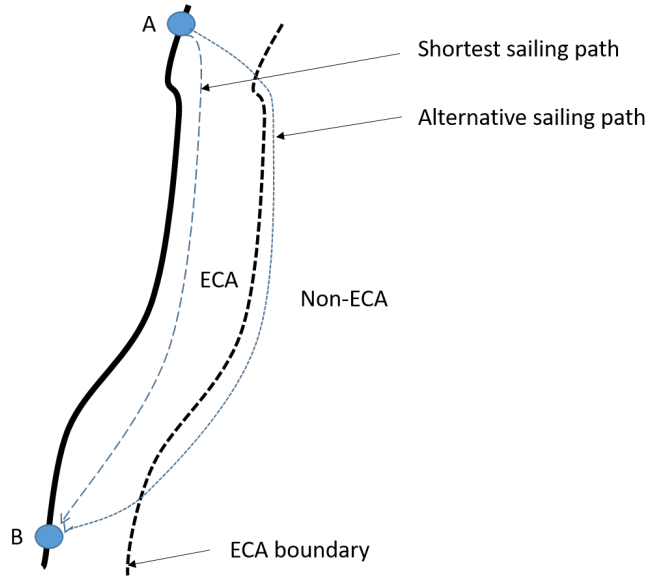


Figure 1: Illustration of alternative sailing paths with ECAs

In this case, we can include in our model a discrete set of alternative sailing paths P_{ij} for the arc (i, j) . For each of these paths $p \in P_{ij}$, we let the distance within and outside the ECA be defined as D_{pij}^{IN} and D_{pij}^{OUT} , respectively. For the shortest sailing path in Figure 1, $D_{pij}^{OUT} = 0$, while D_{pij}^{IN} for the alternative sailing path will be the sum of the distances for the stretches within the ECA when leaving port A and before entering port B. Similarly, we can define a variable for the speed within and outside ECA as s_{pijk}^{IN} and s_{pijk}^{OUT} , respectively, for each path p between the nodes i and j for vessel k . Furthermore, the fuel consumption $C(s_{ijk})$ as a function of speed defined in Section 1.3.2 should now be split into two separate ones $C^{IN}(s_{pijk}^{IN})$ and $C^{OUT}(s_{pijk}^{OUT})$, defining the sailing cost functions for path p between nodes i and j for vessel k within and outside the ECAs, respectively. We also need to introduce an additional flow variable x_{pijk}^E

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which takes the value 1 if vessel k sails along path p between nodes i and j , and 0 otherwise.

The above additional notation is summarized in Table 1.3.3.

Parameters	
D_{pij}^{IN}	Distance within ECAs for sailing path p between nodes i and j
D_{pij}^{OUT}	Distance outside ECAs for sailing path p between nodes i and j
Sets	
P_{ij}	Set of alternative sailing paths between nodes i and j
Decision variables	
s_{pijk}^{IN}	Sailing speed for vessel k within ECAs along path p between nodes i and j
s_{pijk}^{OUT}	Sailing speed for vessel k outside ECAs along path p between nodes i and j
x_{pijk}^E	Binary variable which takes the value 1 if vessel k sails along path p between nodes i and j , and 0 otherwise.
Functions	
$C^{IN}(s_{pijk}^{IN})$	Sailing cost for vessel k when using speed s_{pijk}^{IN} within ECAs along path p between nodes i and j
$C^{OUT}(s_{pijk}^{OUT})$	Sailing cost for vessel k when using speed s_{pijk}^{OUT} outside ECAs along path p between nodes i and j

Table 1.3.3: Additional notation for the ECA extension

With this new notation, we can update the third sailing cost term of our objective function (1.37) as follows.

$$- \sum_{k \in K} \sum_{(i,j) \in A_k} \sum_{p \in P_{ij}} \left(C^{IN}(s_{pijk}^{IN}) x_{pijk}^E + C^{OUT}(s_{pijk}^{OUT}) x_{pijk}^E \right) \quad (1.41)$$

We also need to update the time constraints (1.38) and (1.39), as well as the speed limit constraints (1.40), as follows.

$$x_{pijk}^E \left(t_{ik} + T_{ik}^P + D_{pij}^{IN}/s_{pijk}^{IN} + D_{pij}^{OUT}/s_{pijk}^{OUT} - t_{jk} \right) \leq 0 \quad k \in K, (i,j) \in A_k, p \in P_{ij} \quad (1.42)$$

$$t_{ik} + T_{ik}^P + D_{i,n+i}/\bar{S}_k - t_{n+i,k} \leq 0 \quad k \in K, i \in V_k^P \quad (1.43)$$

$$\underline{S}_k \leq s_{pijk}^{IN}, s_{pijk}^{OUT} \leq \bar{S}_k \quad k \in K, (i,j) \in A_k, p \in P_{ij} \quad (1.44)$$

Next, we need to make sure that we pick at most one of the candidate sailing paths between any pair of nodes, which is ensured by the following constraints.

$$\sum_{p \in P_{ij}} x_{pijk}^E = x_{ijk} \quad k \in K, (i,j) \in A_k \quad (1.45)$$

$$x_{pijk}^E \in \{0, 1\} \quad k \in K, (i, j) \in A_k, p \in P_{ij} \quad (1.46)$$

To our knowledge, no studies have yet included the above ECA considerations in any tramp or industrial ship routing and scheduling problems. However, this has been done in the case when the routes are given (e.g., in liner shipping). Sheng et al. (2019) considered speed optimization within and outside ECAs along given routes when the sailing paths between the ports along each route were given. Fagerholt et al. (2015) and Zhen et al. (2020) also included the decision of selecting a path from a discrete set of candidate paths on each sailing leg (in a similar way as shown above).

Note also that the above modeling approach can be used in other situations with alternative sailing paths. As an example, for sailings between Asia and Europe a vessel may go 1) through the Suez Canal, 2) around the Cape of Good Hope in South Africa, or 3) via the Northern Sea Route (Li et al., 2022).

1.3.4 Other important extensions

In the following, we briefly present two additional significant extensions to the TSRSP, namely flexible cargo sizes and split loads. However, it is important to note that tramp and industrial ship routing problems manifest in various forms across different shipping segments. Hence, numerous practical extensions exist beyond those outlined in this chapter. These include considerations such as voyage separation requirements (Branchini et al., 2015; Vilhelmsen et al., 2017), dynamic and/or stochastic versions of the TSRSP (Hwang et al., 2008; Tirado et al., 2013), and batch acceptance of cargoes (Andersson et al., 2011b; Wu et al., 2021). Another important aspect of tramp shipping is cargo trade, where the routing of vessels may depend on (stochastic) trade market aspects, such as demand and supply in various regions and price speculation. This introduces additional uncertainty to the TSRSP. Furthermore, there are Maritime Inventory Routing Problems (MIRPs) that integrate vessel routing and scheduling with inventory management in the ports. For a comprehensive overview of MIRPs, we refer to the recent survey by Fagerholt et al. (2023).

Flexible cargo sizes

In many TSRSPs in bulk shipping, it is common that the ship operators engage in long-term contracts with their customers where cargoes are transported between given ports on a recurrent basis. In these situations, the exact size of each cargo is not important, and the contracts often state that the size of the cargoes is flexible within some limits. For example, a contract might state that a given cargo should be between 27,000 and 33,000 tonnes instead of exactly 30,000 tonnes. This means that the ship operator can choose to transport any amount within this interval. In this case, the cargo sizes become decision variables instead of input parameters to the TSRSP. By optimizing the cargo sizes, it might be possible to utilize this flexibility to increase the vessel fleet utilization, e.g., to transport more spot cargoes with increased revenues and profits as a result.

Determining the optimal cargo sizes can, similarly to e.g., speed optimization, be done locally within each vessel route, as these quantity decisions within one route do not affect the other routes. Brønmo et al. (2007) solved the TSRSP with flexible cargo sizes with a set partitioning model where the candidate vessel routes are generated a priori and optimized with respect to the cargo sizes. Brønmo et al. (2010) extended this work with a branch-and-price algorithm where

the cargo quantities are discretized, while Korsvik and Fagerholt (2010) proposed a tabu search algorithm embedding a specialized heuristic for deciding the optimal cargo sizes.

Split loads

In most versions of the TSRSP, each cargo should be transported by only one vessel. However, similar to split deliveries in the Vehicle Routing Problem (Dror et al., 1994), one can also in the TSRSP allow for more than one vessel to transport the full volume of a cargo. As for the flexible cargo sizes, this can be utilized to increase fleet utilization and improve profits. In contrast to the extensions with speed optimization and flexible cargo sizes, the decisions on how to split the loads cannot be handled locally within each route. This is because if one decides to split a given cargo, at least two vessels must ensure that the full volume of the cargo is transported.

Different split-load versions of the TSRSP have been studied. Some solution methods are based on alternative set partitioning models (Andersson et al., 2011a; Stålhane et al., 2012; Nishi and Izuno, 2014; Hennig et al., 2015), while others are based on search heuristics (Korsvik et al., 2011; Lee and Kim, 2015).

1.4 Concluding remarks and future trends

The Tramp Ship Routing and Scheduling Problem (TSRSP) has been examined with a focus on various extensions to address real-world complexities in tramp and industrial shipping. Key extensions discussed include considerations for bunker optimization, speed optimization, routing with Emission Control Areas (ECAs), and other significant aspects such as flexible cargo sizes and split loads.

This chapter has provided insights into the models designed to tackle these planning problems. Both a vessel flow and a set partitioning model are presented for the basic TSRSP, and extensions to the vessel flow model are formulated for the bunker optimization, speed optimization, and routing with ECAs, each posing unique modeling challenges. Solution methods for the various planning problems and underlying models are briefly discussed. In summary, the TSRSP is a dynamic and evolving problem with multiple facets, and researchers continue to explore innovative approaches to address its complexities and practical implications in the context of tramp and industrial shipping.

Looking ahead, the operations research field in tramp ship routing and scheduling is ready for further advancements. Future research endeavors are likely to focus on refining existing models, developing even more sophisticated solution methods, and addressing emerging challenges in the dynamic landscape of maritime transportation. Collaborative efforts between academia and industry are essential to ensure that research outcomes are not only theoretically sound but also practically applicable.

When considering future trends in tramp and industrial ship routing, it is important to recognize the ongoing changes in the shipping industry. A complex transformation is happening, influenced by advancements in technology, environmental requirements, and geopolitical shifts. This combination is set to alter the conventional approaches to ship routing and scheduling.

The potential increased use of autonomous ships is a significant shift that poses challenges and opportunities. As autonomous technologies advance, the role of human intervention in routing decisions may change, necessitating a reevaluation of routing models to accommodate this transformative trend.

The advent of new fuel types, driven by environmental concerns and regulatory frameworks, has already begun to influence vessel operations. The industry's response to these changes will necessitate a recalibration of routing strategies, considering the fuel efficiency and environmental impact of different routes. Furthermore, new ECAs, such as those expected in the Mediterranean by 2025, will introduce additional constraints and considerations for ship routing. Adhering to strict emission regulations may require route adjustments, impacting the overall efficiency and cost-effectiveness of shipping operations.

Simultaneously, the observed decrease in globalization, driven by geopolitical shifts, economic considerations, and focus on emissions, has profound implications for the shipping landscape. We may witness a shift towards shorter-range transportation using small, adaptable vessels, diverging from the current trend of long-distance transport with mega-vessels. Routes and schedules will likely undergo adjustments to align with evolving trade patterns and regionalization. This shift may result in the exploration of alternative routes, necessitating a reevaluation of existing models to optimize for changing trade dynamics.

Security concerns in an uncertain world are expected to play a crucial role in shaping shipping's future. Routing decisions may need to consider security, leading to potential adjustments based on geopolitical developments. Ensuring the safety of cargo, crew, and vessels may become an integral aspect of routing optimization.

Moreover, melting sea ice, e.g., in the northeast passage, presents both challenges and opportunities. While it opens up new, shorter routes, the dynamic and unpredictable nature of ice melt necessitates careful consideration in routing decisions. Minimizing time spent by ice-classified vessels outside ice-filled areas will be crucial, necessitating new routing strategies.

In summary, the future of tramp and industrial ship routing will be influenced by technological, environmental, and geopolitical trends. Adaptability, innovation, and a holistic approach to routing will be crucial as the industry navigates these dynamic shifts in the coming years.

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